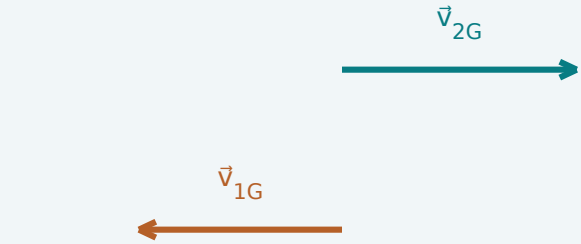
## Concept questions • 1. Two cars in opposite directions

**Question**

Car 1 is driving at 45 km/h [W] with respect to the road. Car 2 is driving at 52 km/h [E] with respect to the road. What is the velocity of car 2 with respect to car 1?

**Original velocities relative to the road**




$\vec{v}_{2G}$: 52 km/h [E]

$\vec{v}_{1G}$: 45 km/h [W]

+x east; −x west

Given velocities before reversing an arrow.

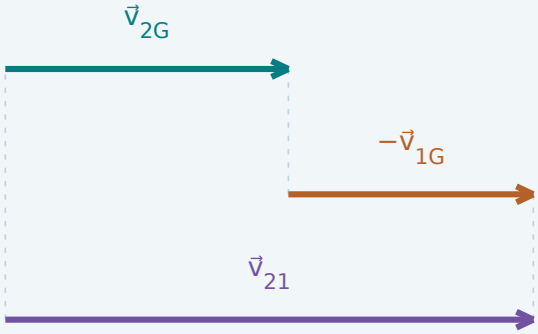
**1. Reverse the observer's velocity.** Car 1 is the observer. Its velocity relative to the road is west, so the road's velocity relative to car 1 is east with the same magnitude.

$$\vec{v}_{21} = \vec{v}_{2G} - \vec{v}_{1G}$$

$$-\vec{v}_{1G} = 45 \text{ km/h [E]}$$

$$\vec{v}_{21} = \vec{v}_{2G} + (-\vec{v}_{1G})$$

**Add car 2's velocity and the opposite of car 1's**




+x east; −x west; dotted guides align the arrow ends

$\vec{v}_{21} = \vec{v}_{2G} + (-\vec{v}_{1G})$

$\vec{v}_{2G}$: 52 km/h [E]

$-\vec{v}_{1G}$: 45 km/h [E]

$\vec{v}_{21}$: 97 km/h [E]

**2. Add the eastward components.** Both arrows in the final sum point east. The result runs from the first tail to the last tip.

$v_{21,x} = (+52) + (+45) = +97$ km/h

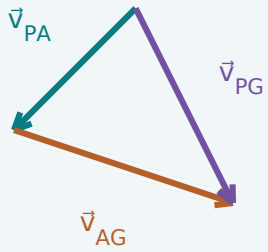
**Final answer: 97 km/h [E] relative to car 1.**

## Concept questions • 2. Identify plane velocity through the air

**Question**

If the wind is represented by the arrow pointing south-east, which arrow represents the velocity of the plane relative to the air?

**Plane through air + wind = plane over ground**




$\vec{v}_{PA}$: Plane relative to air
down-left arrow

$\vec{v}_{AG}$: Air relative to ground
southeast wind

$\vec{v}_{PG}$: Plane relative to ground
first tail → final tip

+x east; −x west; +y north; −y south

$$\vec{v}_{PG} = \vec{v}_{PA} + \vec{v}_{AG}$$

**1. Follow the joined arrows.** The southeast arrow is wind: air moving relative to ground. The arrow ending where the wind begins is the plane moving relative to air.

$$\vec{v}_{PG} = \vec{v}_{PA} + \vec{v}_{AG}$$

**2. Identify the answer.** The plane/air arrow points down and left. Adding the southeast wind arrow gives the plane's ground-velocity arrow.

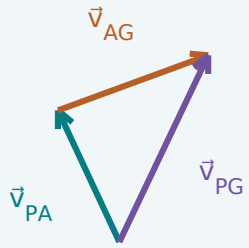
**Final answer: The down-left arrow from the top point to the left point is the plane's velocity relative to the air.**

## Concept questions • 3. Draw the wind vector

**Question**

A pilot aims [N25°W] with an airspeed of 245 km/h in order to achieve the groundspeed shown below. Draw a vector diagram to show the direction of the wind.

### Original velocity relationship




$\vec{v}_{PA}$: 245 km/h [N25°W]

$\vec{v}_{AG}$: Unknown wind
air tip → ground tip

$\vec{v}_{PG}$: 350 km/h
direction shown in worksheet

+x east; −x west; +y north; −y south

$\vec{v}_{PG} = \vec{v}_{PA} + \vec{v}_{AG}$ (schematic)

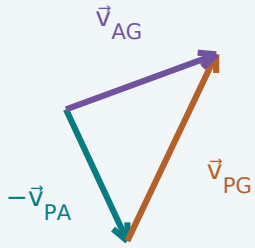
**1. Write the wind as a difference.** The wind is the arrow that takes the plane/air tip to the plane/ground tip.

$$\vec{v}_{AG} = \vec{v}_{PG} - \vec{v}_{PA}$$

**2. Replace subtraction with the opposite velocity.** Reverse the plane/air arrow: [N25°W] becomes [S25°E]. Place the ground-velocity arrow at the reversed arrow's tip, as on your whiteboard.

$$\vec{v}_{AG} = (-\vec{v}_{PA}) + \vec{v}_{PG}$$

**Reversed plane/air arrow + plane/ground arrow**




$-\vec{v}_{PA}$: 245 km/h [S25°E]

$\vec{v}_{PG}$: 350 km/h
copy the given ground arrow

$\vec{v}_{AG}$: Wind points northeast

+x east; −x west; +y north; −y south

$\vec{v}_{AG} = (-\vec{v}_{PA}) + \vec{v}_{PG}$ (schematic)

**3. Read the resultant.** Draw the wind from the first tail to the final tip. It points northeast. The worksheet does not give a numerical ground bearing, so an exact wind bearing cannot be calculated.

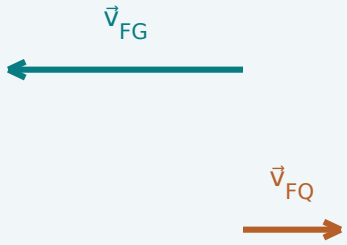
**Final answer: Draw $\vec{v}_{AG}$ from the air arrow’s tip to the ground arrow’s tip; it points roughly northeast.**

## Concept questions • 4. Football seen by a running quarterback

**Question**

A football is thrown through the air at 6.0 m/s [S], but a quarterback running down the field sees it moving at 2.5 m/s [N]. Determine the speed and direction the quarterback is running.

### Original football velocities




$\vec{v}_{FG}$: 6.0 m/s [S]

$\vec{v}_{FQ}$: 2.5 m/s [N]

+y north →; −y south ← (axis drawn horizontally)

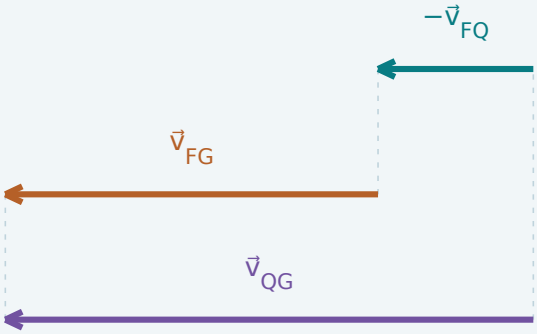
Given velocities before reversing an arrow.

**1. Use the opposite frame velocity.** The ball moves north relative to the quarterback. Therefore the quarterback moves south relative to the ball. Add that southward velocity to the ball's southward ground velocity.

$$\vec{v}_{QG} = \vec{v}_{FG} - \vec{v}_{FQ}$$

$$\vec{v}_{QG} = (-\vec{v}_{FQ}) + \vec{v}_{FG}$$

**Opposite football/QB velocity + football/ground velocity**




$-\vec{v}_{FQ}$: 2.5 m/s [S]

$\vec{v}_{FG}$: 6.0 m/s [S]

$\vec{v}_{QG}$: 8.5 m/s [S]

+y north →; −y south ←; arrow lanes are offset

$\vec{v}_{QG} = (-\vec{v}_{FQ}) + \vec{v}_{FG}$

**2. Add on a north-positive axis.** Both arrows in the new sum point south, so both components are negative.

$v_{QG,y} = (-2.5) + (-6.0) = -8.5$ m/s

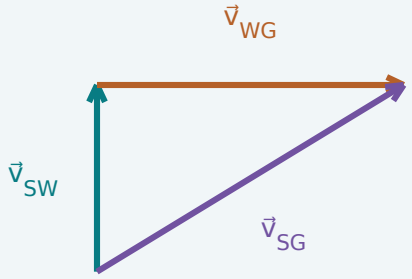
**Final answer: The quarterback runs at 8.5 m/s [S].**

## Worked examples • Example 1. Swimmer and river current

**Question**

A girl tries swimming across a river. The river is flowing at 2.3 m/s [E]. The girl swims at 1.4 m/s [N] with respect to the water. What is the girl's velocity with respect to the river bank (the shore)? Draw a labelled vector diagram and use geometry to solve.

**Swimmer/water + water/shore**




$\vec{v}_{SW}$: 1.4 m/s [N]

$\vec{v}_{WG}$: 2.3 m/s [E]

$\vec{v}_{SG}$: 2.69 m/s [N58.67°E] before final rounding

+x east; −x west; +y north; −y south

$\vec{v}_{SG} = \vec{v}_{SW} + \vec{v}_{WG}$

**1. Name the two motions.** A shore observer sees the water carry the swimmer east while she swims north through that water. Each motion is measured in a different frame; adding the two arrows connects swimmer to shore.

$\vec{v}_{SG} = \vec{v}_{SW} + \vec{v}_{WG}$.

**2. Use the right triangle for speed.** The east and north components are perpendicular. Their lengths form the legs, so Pythagoras gives the magnitude of the diagonal shore velocity.

$|\vec{v}_{SG}| = \sqrt{[(2.3)^2+(1.4)^2]} = 2.69$ m/s.

**3. State the bearing.** Measure θ from north toward east. The east leg is opposite θ and the north leg is adjacent.

$\tan\theta = 2.3/1.4 \Rightarrow \theta = 58.67°$.

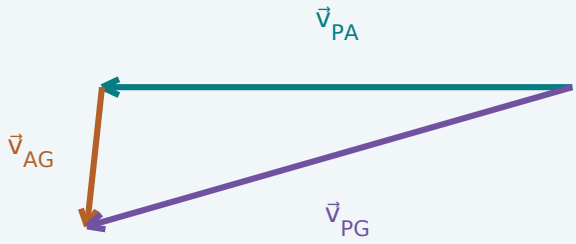
**Final answer: 2.7 m/s [N59°E] relative to the shore.**

## Worked examples • Example 2. Find the wind from two plane velocities

**Question**

A plane is flying with an airspeed of 340 km/h and a heading of west. The plane’s velocity over the ground is 365 km/h [W16°S]. What is the wind velocity? Draw a labelled vector diagram and use geometry or components to solve.

### Original velocity relationship




$\vec{v}_{PA}$: 340 km/h [W]

$\vec{v}_{AG}$: Unknown wind

$\vec{v}_{PG}$: 365 km/h [W16°S]

+x east; −x west; +y north; −y south

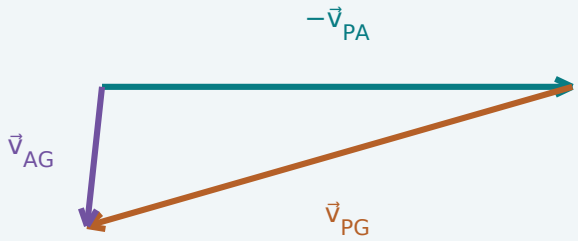
$$\vec{v}_{PG} = \vec{v}_{PA} + \vec{v}_{AG}$$

**1. Solve for wind using addition.** Reverse the plane/air vector from west to east. Add the ground vector at its tip. The resultant is wind.

$$\vec{v}_{AG} = \vec{v}_{PG} - \vec{v}_{PA}$$

$$\vec{v}_{AG} = (-\vec{v}_{PA}) + \vec{v}_{PG}$$

**Add the opposite air velocity to the ground velocity**




$-\vec{v}_{PA}$: 340 km/h [E]

$\vec{v}_{PG}$: 365 km/h [W16°S]

$\vec{v}_{AG}$: Mostly south, slightly west

+x east; −x west; +y north; −y south

$\vec{v}_{AG} = (-\vec{v}_{PA}) + \vec{v}_{PG}$

**2. Resolve the known ground velocity.** Use east +x and north +y. [W16°S] has both negative components; the 340 km/h air velocity has only a west component.

$v_{PG,x} = -365 \cos 16° = -350.86$ km/h; $v_{PG,y} = -365 \sin 16° = -100.61$ km/h.

$(v_{PA,x}, v_{PA,y}) = (-340, 0)$ km/h.

**3. Add the components of the two pictured arrows.** The reversed air velocity is (+340, 0) km/h. The ground velocity has west and south components. Add corresponding components; keep the trig expressions unrounded.

$v_{AG,x} = (+340) + (-365 \cos 16°) = -10.86$ km/h

$v_{AG,y} = 0 + (-365 \sin 16°) = -100.61$ km/h

**4. Find its speed and bearing.** The magnitude follows from the two perpendicular components. Measure the angle west of south because south is the larger component.

$|\vec{v}_{AG}| = \sqrt{[(365 \cos 16° - 340)^2 + (365 \sin 16°)^2]} = 101.19$ km/h.

$\theta = \tan^{-1}[(365 \cos 16° - 340)/(365 \sin 16°)] = 6.16°$ west of south.

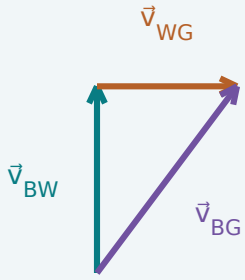
**Final answer: 101 km/h [S6.2°W] relative to ground (340 km/h treated as three significant figures).**

## Problems to solve • 1. Boat aimed straight across

**Question**

A boat crosses a river of width 160 m in which the current has a uniform speed of 1.50 m/s. The pilot maintains a bearing perpendicular to the river and a throttle setting to give a constant speed of 2.00 m/s relative to the water. (a) What is the speed of the boat relative to a stationary shore observer? (b) How far downstream from the initial position is the boat when it reaches the opposite shore?

### Boat across water + downstream current




$\vec{v}_{BW}$: 2.00 m/s across

$\vec{v}_{WG}$: 1.50 m/s downstream

$\vec{v}_{BG}$: 2.50 m/s diagonally relative to shore

+x downstream; −x upstream; +y across; −y back

$\vec{v}_{BG} = \vec{v}_{BW} + \vec{v}_{WG}$

**1. Add the velocities in the shore frame.** The boat points straight across relative to the water, but the moving water also carries it downstream. The shore velocity is the diagonal sum of these perpendicular motions.

$\vec{v}_{BG} = \vec{v}_{BW} + \vec{v}_{WG}$.

$|\vec{v}_{BG}| = \sqrt{[(2.00)^2+(1.50)^2]} = 2.50$ m/s.

**2. Give the ground direction.** The downstream component is opposite an angle measured from across; the 2.00 m/s component is adjacent.

$\theta = \tan^{-1}(1.50/2.00) = 36.87°$ downstream of across.

**3. Calculate crossing time from the across motion only.** The river is 160 m wide. The current changes the landing point but adds no across speed, so the 2.00 m/s across component closes that width.

$t = 160/(2.00) = 80.00$ s.

**4. Calculate drift during that same time.** The water carries the boat downstream for all 80.00 s. Multiply the downstream component by the crossing time.

$d_{down} = (1.50)(160/2.00) = 120$ m.

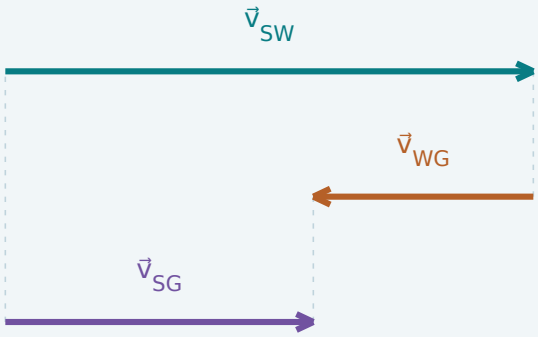
**Final answer: (a) 2.50 m/s (ground direction: across 36.9° downstream). (b) $1.2 \times 10^2$ m downstream (160 m treated as two significant figures).**

## Problems to solve • 2. Swim upstream and return

**Question**

A river has a steady speed of 0.500 m/s. A student swims upstream a distance of 1.00 km and the swims back to the original starting point. If the student can swim at a speed of 1.20 m/s in still water, how long does the trip take? Compare this with the time the trip would take if the water were still.

### Upstream leg: add the opposing current




+x upstream; −x downstream

$\vec{v}_{SG} = \vec{v}_{SW} + \vec{v}_{WG}$

$\vec{v}_{SW}$: 1.20 m/s upstream

$\vec{v}_{WG}$: 0.500 m/s downstream

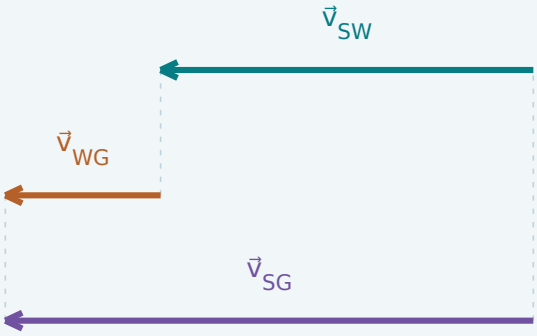
$\vec{v}_{SG}$: 0.700 m/s upstream

**1. Use the shore speeds for travel times.** The 1.00 km distances are measured along the shore. On the first leg the current subtracts from the swimmer's upstream motion; on return both velocities point downstream.

$u_{up} = 1.20 - 0.500 = 0.700$ m/s.

$u_{down} = 1.20 + 0.500 = 1.70$ m/s.

**Return leg: keep the same positive direction**




$\vec{v}_{SW}$: 1.20 m/s downstream

$\vec{v}_{WG}$: 0.500 m/s downstream

$\vec{v}_{SG}$: 1.70 m/s downstream

+x upstream; −x downstream

$\vec{v}_{SG} = \vec{v}_{SW} + \vec{v}_{WG}$

**2. Compute each leg separately.** There are two 1000 m legs at different speeds. Dividing the entire 2000 m by one of those speeds would give the wrong time.

$t_{up} = 1000/(1.20-0.500) = 1428.57$ s.

$t_{down} = 1000/(1.20+0.500) = 588.24$ s.

$t_{current} = 1428.57 + 588.24 = 2016.81$ s.

**3. Compare with still water.** Without a current, both 1000 m legs are at 1.20 m/s. The round-trip displacement is zero, but the distance is 2000 m; use distance for this travel-time comparison.

$t_{still} = 2000/1.20 = 1666.67$ s.

$\Delta t = t_{current} - t_{still} = 350.14$ s $\approx 5.84$ min.

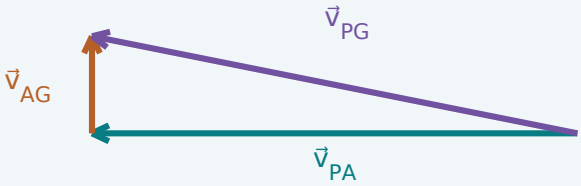
**Final answer: With current: $2.02 \times 10^3$ s. Still water: $1.67 \times 10^3$ s. Extra time: $3.50 \times 10^2$ s (5.84 min).**

## Problems to solve • 3. Plane heading west in north wind

**Question**

The pilot of an airplane notes that the compass indicates a heading due west. The plane's airspeed (speed relative to the air) is 150 km/h. If there is a wind of 30.0 km/h toward the north, find the velocity of the plane with respect to the ground.

**Westward plane/air velocity + northward wind**




$\vec{v}_{PA}$: 150 km/h [W]

$\vec{v}_{AG}$: 30.0 km/h [N]

$\vec{v}_{PG}$: Ground track points northwest

+x east; −x west; +y north; −y south

$\vec{v}_{PG} = \vec{v}_{PA} + \vec{v}_{AG}$

**1. Add the motions, not just their speeds.** Airspeed and wind point in perpendicular directions. The plane moves west through the air; that same air moves north over the ground. Joining these arrows gives the ground velocity.

$\vec{v}_{PG} = \vec{v}_{PA} + \vec{v}_{AG}$.

$(v_{PG,x}, v_{PG,y}) = (-150, +30.0)$ km/h.

**2. Find the ground speed.** The west and north components form a right triangle; add their squares, then take the square root.

$|\vec{v}_{PG}| = \sqrt{[(150)^2 + (30.0)^2]} = 152.97$ km/h.

**3. Find the direction.** Measure the angle from west toward north. The north leg is opposite that angle.

$\theta = \tan^{-1}(30.0/150) = 11.31°$ north of west.

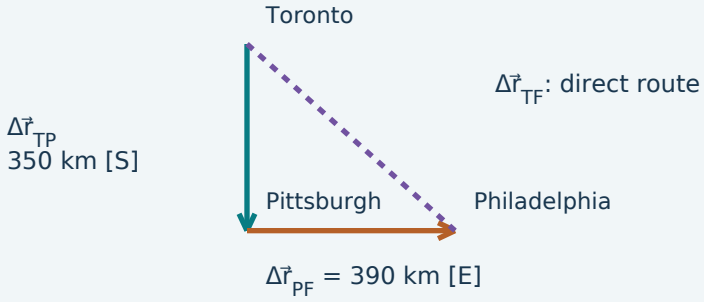
**Final answer: 153 km/h [W11.3°N] relative to ground (150 km/h treated as three significant figures).**

## Problems to solve • 4. Two routes to Philadelphia

**Question**

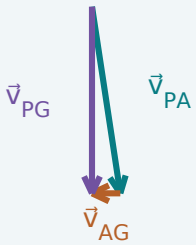
Two planes fly from Toronto to Philadelphia. Plane A flies via Pittsburgh whereas passengers on plane B have a direct flight. Pittsburgh is 350 km due south of Toronto and 390 km due west of Philadelphia. The airspeed of both planes is 400.0 km/h and a steady wind is blowing from the east at 60.0 km/h. (a) What direction must the pilot point the plane flying from Toronto to Pittsburgh? Include a vector diagram of velocities. (b) How long will the entire flight take for plane A assuming a 0.50-h layover in Pittsburgh? (c) How much time must the pilot of plane B wait before leaving Toronto if she is to arrive in Philadelphia at the same time plane A arrives?

**Ground routes from Toronto**




Positions measured relative to shore / ground.

**Plane A: first leg to Pittsburgh**




$\vec{v}_{PA}$: 400.0 km/h [S8.63°E]

$\vec{v}_{AG}$: 60.0 km/h [W]

$\vec{v}_{PG}$: 395.47 km/h [S]
before final rounding

+x east; −x west; +y north; −y south

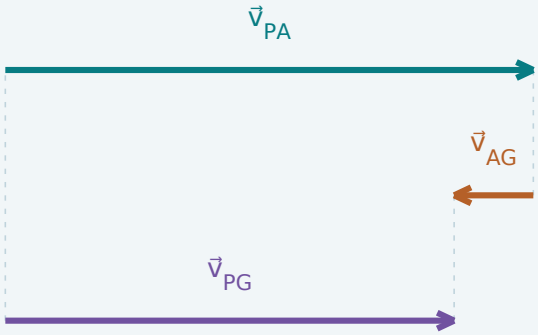
$\vec{v}_{PG} = \vec{v}_{PA} + \vec{v}_{AG}$

**(a) Cancel wind on the first leg.** The target Pittsburgh is directly south, so the plane needs an eastward air-velocity component equal to the westward wind. The remaining component carries it south.

$400.0 \sin\theta = 60.0 \Rightarrow \theta = 8.63°$ east of south.

$u_{south} = \sqrt{[(400.0)^2 - (60.0)^2]} = 395.47$ km/h.

**Plane A: second leg to Philadelphia**




$\vec{v}_{PA}$: 400.0 km/h [E]

$\vec{v}_{AG}$: 60.0 km/h [W]

$\vec{v}_{PG}$: 340.0 km/h [E]

+x east; −x west

$\vec{v}_{PG} = \vec{v}_{PA} + \vec{v}_{AG}$

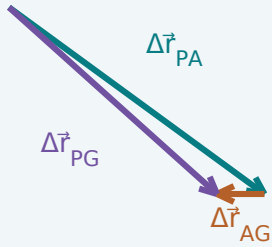
**(b) Time each part of A's itinerary.** Use the ground speed along each route leg. On the eastbound leg, the 60.0 km/h west wind opposes the full 400.0 km/h east airspeed. Add the 0.50 h layover only after both flight times.

$t_{south} = 350/\sqrt{[(400.0)^2 - (60.0)^2]} = 0.89$ h.

$u_{east} = 400.0 - 60.0 = 340.0$ km/h; $t_{east} = 390/340.0 = 1.15$ h.

$t_A = 350/\sqrt{[(400.0)^2 - (60.0)^2]} + 0.50 + 390/340.0 = 2.53$ h.

**Plane B: displacement through air + wind displacement**




$\Delta\vec{r}_{PA}$: (390 + 60.0t) km [E] and 350 km [S]

$\Delta\vec{r}_{AG}$: 60.0t km [W]

$\Delta\vec{r}_{PG}$: 390 km [E], 350 km [S]

+x east; −x west; +y north; −y south

$\Delta\vec{r}_{PG} = \Delta\vec{r}_{PA} + \Delta\vec{r}_{AG}$

**(c) Work in the ground frame for B's route.** B must move 390 km east and 350 km south. In t hours the wind pushes it 60.0t km west, so its eastward air displacement must be 390+60.0t km. Its air-distance magnitude is 400.0t km.

$(390+60.0t)^2 + (350)^2 = (400.0t)^2$.

$152100 + 46800t + 3600t^2 + 122500 = 160000t^2$.

$156400t^2 - 46800t - 274600 = 0$.

$t_B = [46800 + \sqrt{(46800^2 + 4(156400)(274600))}]/[2(156400)] = 1.48$ h.

**Compare the arrival clocks.** Plane A's total interval includes its stop. Plane B flies directly, so it must leave later by the difference between those intervals.

wait $= t_A - t_B = 2.53 - 1.48 \approx 1.05$ h $\approx 63$ min.

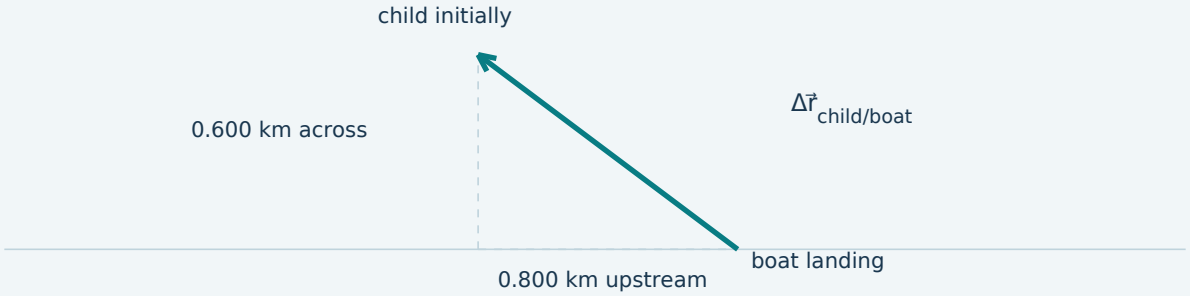
**Final answer: (a) Aim [S8.63°E]. (b) Plane A takes 2.5 h including the stop. (c) Plane B waits about 1.0 h (about 63 min).**

## Problems to solve • 5. Rescue a child in a river

**Question**

A child in danger of drowning in a river is being carried downstream by a current that has a speed of 2.50 km/h. The child is 0.600 km from shore and 0.800 km upstream of a boat landing when a rescue boat sets out. (a) If the boat proceeds at its maximum speed of 20.0 km/h relative to the water, in what direction should the pilot aim the boat in order to intercept the child? (b) What will be the velocity of the boat with respect to the shore? (c) How long will it take the boat to reach the child?

**Initial boat-to-child separation**




Positions measured relative to shore / ground.

**(a) Change to the water frame to aim.** The current moves the child and boat equally. Relative to the water, the child stays 0.800 km upstream and 0.600 km across from the landing. Aim directly at that location in the water frame.

$d_{BW} = \sqrt{[(0.800)^2+(0.600)^2]} = 1.00$ km.

$\theta = \tan^{-1}(0.600/0.800) = 36.87°$ from the upstream shoreline toward across.

**(c) Find how long the boat needs.** The boat covers the 1.00 km water-frame separation at 20.0 km/h. We can find interception time before its shore velocity because both boat and child drift together.

$t = 1.00/20.0 = 0.0500$ h $= 3.00$ min.

**(b) Resolve the boat's water-relative velocity.** The aiming triangle has upstream and across fractions 0.800/1.00 and 0.600/1.00. Choose downstream +x, across +y. Its upstream component is negative.

$v_{BW,x} = -20.0(0.800/1.00) = -16.0$ km/h.

$v_{BW,y} = +20.0(0.600/1.00) = +12.0$ km/h.

**Boat/water + current = boat/shore**




$\vec{v}_{BW}$: 16.0 km/h upstream
12.0 km/h across

$\vec{v}_{WG}$: 2.50 km/h downstream

$\vec{v}_{BG}$: 13.5 km/h upstream
12.0 km/h across

+x downstream; −x upstream; +y across; −y back

$\vec{v}_{BG} = \vec{v}_{BW} + \vec{v}_{WG}$

**Add the current and state the shore velocity.** Only the x component changes because the 2.50 km/h current is downstream. The boat still moves upstream over the shore, but less quickly than through the water.

$v_{BG,x} = -16.0+2.50 = -13.5$ km/h; $v_{BG,y} = +12.0$ km/h.

$|\vec{v}_{BG}| = \sqrt{[(13.5)^2+(12.0)^2]} = 18.06$ km/h.

$\varphi = \tan^{-1}(12.0/13.5) = 41.63°$ from upstream toward across.

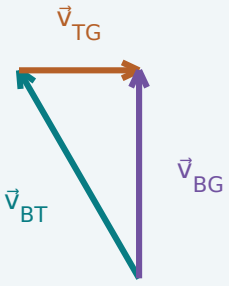
**Final answer: (a) Aim 36.9° from upstream along the shore toward across (or 53.1° upstream from straight across). (b) 18.1 km/h, 41.6° from upstream toward across. (c) 0.0500 h = 3.00 min.**

## Problems to solve • 6. Ball appears to rise vertically

**Question**

A physics student is riding on a flatcar of a train travelling along a straight horizontal track at a constant velocity of 10.0 m/s. The student throws a ball into the air along a path that he judges to make an initial angle of 60.0° with the horizontal and to be in line with the track. The student's teacher, who is standing on the ground nearby, observes the ball to rise vertically. How high does the professor see the ball rise?

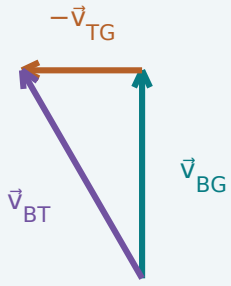
### Original frame relationship




$\vec{v}_{BG} = \vec{v}_{BT} + \vec{v}_{TG}$

+x train-forward; −x backward; +y up; −y down

### Add the opposite velocity




$\vec{v}_{BT} = \vec{v}_{BG} + (-\vec{v}_{TG})$

+x train-forward; −x backward; +y up; −y down

**1. Cancel the horizontal ground motion.** The teacher sees a vertical path, so the ball's horizontal ground component is zero. Reverse the forward train velocity and add it to the ball's ground velocity to get the throw relative to the train.

$\vec{v}_{BT} = \vec{v}_{BG} - \vec{v}_{TG}$

$\vec{v}_{BT} = \vec{v}_{BG} + (-\vec{v}_{TG})$

$v_{BT,x} = 0 + (-10.0) = -10.0$ m/s

**2. Find its upward launch component.** The stated 60.0° is above the backward horizontal. In the train-frame right triangle, the horizontal leg has magnitude 10.0 m/s and the vertical leg is opposite 60.0°. The train adds no vertical velocity.

$\tan 60.0° = v_{BT,y}/10.0 \Rightarrow v_{BG,y} = 10.0 \tan 60.0° = 17.32$ m/s.

**3. Use vertical kinematics for the rise.** At the highest point, vertical velocity is zero. The upward speed is the teacher's launch y component; gravity slows it at 9.81 m/s². Use the unrounded tangent expression.

$0 = [10.0 \tan 60.0°]^2 - 2(9.81)h.$

$h = [10.0 \tan 60.0°]^2/[2(9.81)] = 15.29$ m.

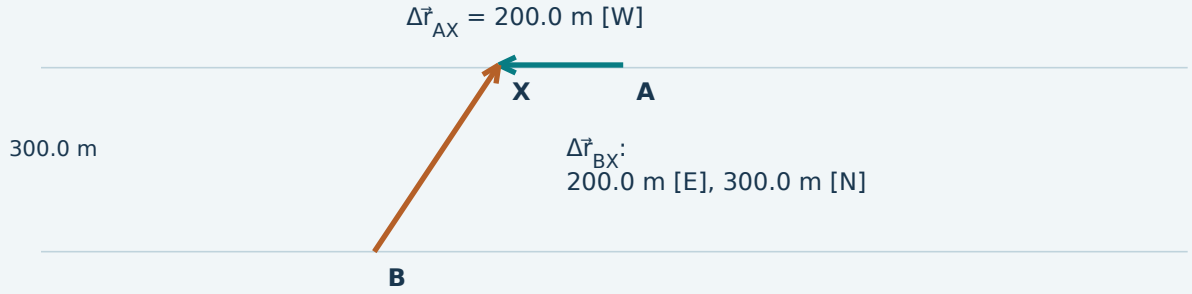
**Final answer: The teacher sees the ball rise 15.3 m above its launch point.**

## Problems to solve • 7. Two canoeists visit X

**Question**

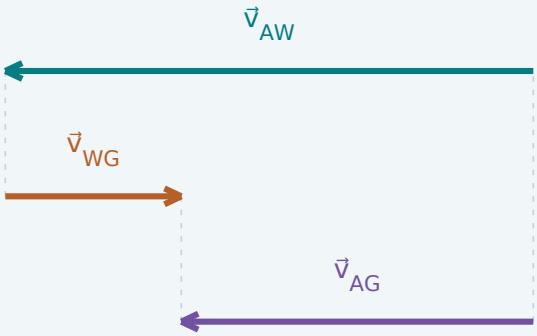
Two canoeists, A and B, live on opposite shores of a 300.0 m wide river that flows east at 0.80 m/s. A lives on the north shore and B lives on the south shore. They both set out to visit a mutual friend X who lives on the north shore at a point 200.0 m upstream from A and 200.0 m downstream from B. Both canoeists can propel their canoes at 2.4 m/s through the water. How much time must canoeist A wait after canoeist B sets out so that they both arrive at X at the same time? Both canoeists make their respective trips by the most direct routes.

**Starting positions and direct routes to X**




Positions measured relative to shore / ground.

**Canoeist A: westward paddling + eastward current**




$\vec{v}_{AW}$: 2.4 m/s [W]

$\vec{v}_{WG}$: 0.80 m/s [E]

$\vec{v}_{AG}$: 1.60 m/s [W]
before final rounding

+x east; −x west

$\vec{v}_{AG} = \vec{v}_{AW} + \vec{v}_{WG}$

**1. Find A's time.** A travels 200.0 m west along the north shore. Paddling west at 2.4 m/s against 0.80 m/s east current gives a 1.60 m/s west ground speed.

$u_{AG} = 2.4 - 0.80 = 1.60$ m/s.

$t_A = 200.0/1.60 = 125.00$ s.

**2. Write B's water-relative velocity as a sum.** B's direct ground route is 200.0 m east and 300.0 m north. Divide each displacement by the same travel time. Then add the opposite of the eastward current: 0.80 m/s west.

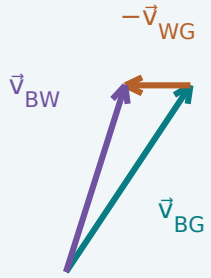
$\vec{v}_{BW} = \vec{v}_{BG} - \vec{v}_{WG}$

$\vec{v}_{BW} = \vec{v}_{BG} + (-\vec{v}_{WG})$

$v_{BG,x} = 200.0/t$; $v_{BG,y} = 300.0/t$

$v_{BW,x} = 200.0/t + (-0.80)$; $v_{BW,y} = 300.0/t + 0$

**Canoeist B: ground velocity + opposite current**




$\vec{v}_{BG}$: (200.0/t, 300.0/t) m/s

$-\vec{v}_{WG}$: 0.80 m/s [W]

$\vec{v}_{BW}$: Through-water speed 2.4 m/s

+x east; −x west; +y north; −y south

$\vec{v}_{BW} = \vec{v}_{BG} + (-\vec{v}_{WG})$

**3. Use B's known paddling speed.** The through-water x and y components form a right triangle with magnitude 2.4 m/s. This single equation contains the unknown crossing time.

$(200.0/t - 0.80)^2 + (300.0/t)^2 = (2.4)^2$.

**Show the algebra for B's time.** Multiply the speed equation by $t^2$ to remove the denominators, then expand and collect powers of t. This uses the water-relative speed, not the ground speed.

$(200.0 - 0.80t)^2 + (300.0)^2 = (2.4t)^2$

$40000 - 320t + 0.64t^2 + 90000 = 5.76t^2$

$130000 - 320t - 5.12t^2 = 0$

$5.12t^2 + 320t - 130000 = 0$

**4. Solve and compare arrival times.** Take the positive root, since elapsed time cannot be negative. It is longer than A's trip, so A can leave later.

$t_B = [-320+\sqrt{(320^2+4(5.12)(130000)}]/[2(5.12)] = 131.13$ s.

$v_{BG,x} = 200.0/t_B \approx 1.53$ m/s; $v_{BG,y} = 300.0/t_B \approx 2.29$ m/s.

wait $= t_B - t_A = 131.13 - 125.00 = 6.13$ s.

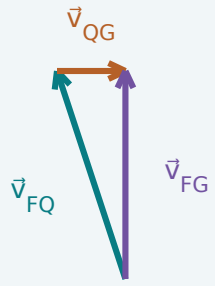
**Final answer: Canoeist A should leave about 6.1 s after B. B's ground route is [E56.3°N]; the worksheet's 72.4° hint does not match the stated distances.**

## Problems to solve • 8. Quarterback aims against his own run

**Question**

A quarterback is running across the field, parallel to the line of scrimmage, at a constant speed of 2.5 m/s, when he spots an open, stationary receiver straight downfield from him (in a line parallel with the sidelines). If he can throw the football at a speed of 8.0 m/s, relative to himself, at what angle, relative to the sidelines, must he throw it in order to hit the receiver? How far downfield was the receiver, if the pass took 3.0 s to reach him?

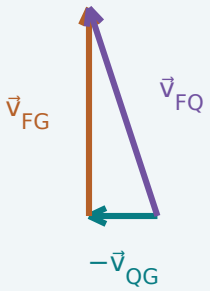
### Original frame relationship




$$\vec{v}_{FG} = \vec{v}_{FQ} + \vec{v}_{QG}$$

+x quarterback's run; −x opposite; +y downfield

### Add the opposite velocity




$$\vec{v}_{FQ} = (-\vec{v}_{QG}) + \vec{v}_{FG}$$

+x quarterback's run; −x opposite; +y downfield

**1. Remove the quarterback's sideways motion.** To reach a stationary receiver directly downfield, the football needs zero sideways ground velocity. Add the opposite of the quarterback's run to that ground velocity to get the throw relative to him.

$$\vec{v}_{FQ} = \vec{v}_{FG} - \vec{v}_{QG}$$

$$\vec{v}_{FQ} = (-\vec{v}_{QG}) + \vec{v}_{FG}$$

$$v_{FQ,x} = (-2.5) + 0 = -2.5 \text{ m/s}$$

**2. Find the aim angle from the sideline direction.** The 8.0 m/s throw is the slanted arrow. Its backward component has magnitude 2.5 m/s, opposite the angle measured from straight downfield.

$\sin\theta = 2.5/8.0 \Rightarrow \theta = 18.21°$ backward from downfield.

**3. Find the downfield speed and distance.** The quarterback contributes no downfield velocity. Use Pythagoras on the throw components, then multiply the downfield speed by the 3.0 s flight time.

$v_{FG,y} = \sqrt{[(8.0)^2 - (2.5)^2]} = 7.60$ m/s.

$d_{downfield} = \sqrt{[(8.0)^2 - (2.5)^2]}(3.0) = 22.80$ m.

**Final answer: Aim 18° backward from the sideline direction; the receiver is 23 m downfield.**